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Carrera Lic. en Ciberdefensa

Álgebra I

Profesor a cargo: Lic. Nelson Monzón

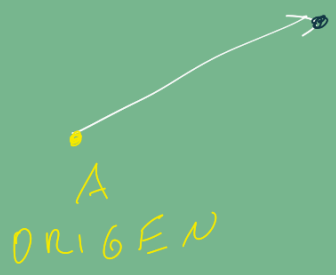
Profesora adjunta: Prof. Maela Lombardi

Año: 2025

Carga horaria: 64

VECTORES

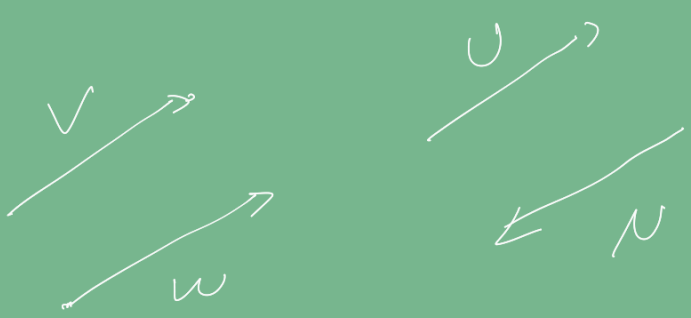
$\overrightarrow{AB} = \vec{v}$

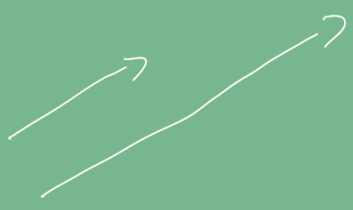


DIRECCIÓN

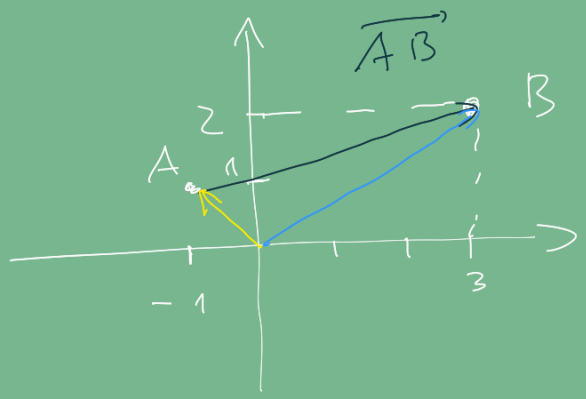
DOS SENTIDOS

VECTORES EQUIVALENTES





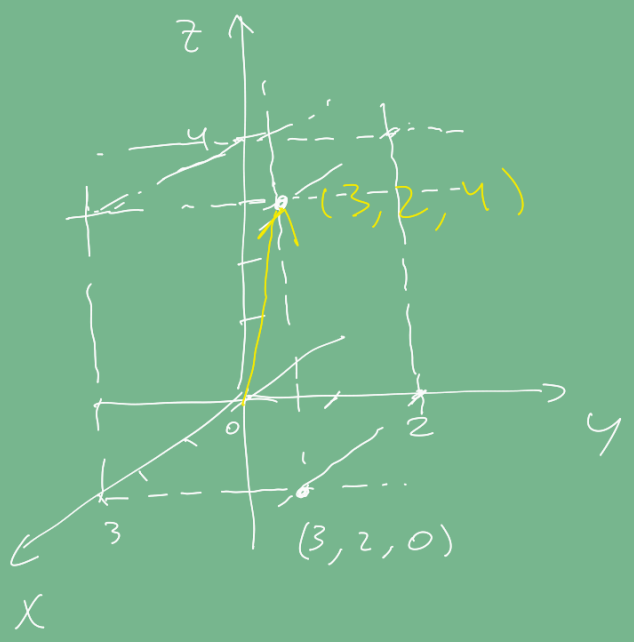
$\mathbb{R}^2 = \{(x,y) : x \in \mathbb{R}, y \in \mathbb{R}\}$



$\vec{A} = (-1,1)$ $\vec{B} = (3,2)$

$(0,0) = O$

$$\mathbb{R}^3 = \{(x,y,z) : x,y,z \in \mathbb{R}\}$$



PROBAR EN GEOGEBRA

$$\mathbb{R}^n = \{\underbrace{(x_1, x_2, \dots, x_n)}_{n\text{-UPLA DE NÚMEROS REALES.}} : x_1, x_2, \dots, x_n \in \mathbb{R}\}$$

$$\vec{O} = \underbrace{(0,0,\dots,0)}_{\text{TAMAÑO } n}$$

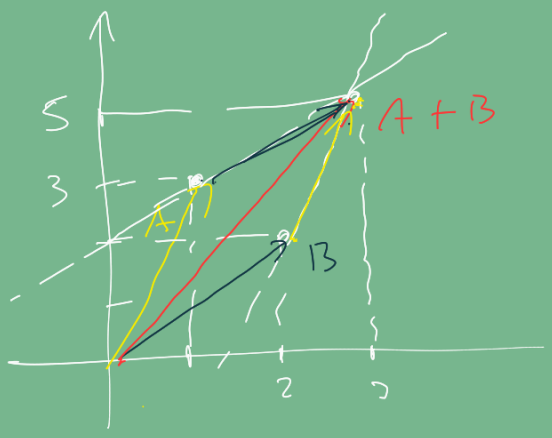
OPERACIONES EN $\mathbb{R}^n$:

SUMA: SI $A = (a_1, a_2, \dots, a_n)$,
$B = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$,

$$A + B \overset{d}{=} (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

ES: $A = (1,3)$; $B = (2,2)$

$$A + B = (1,3) + (2,2) = (3,5)$$



PRODUCTO POR UN ESCALAR:

DADOS $A = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$ Y $\lambda \in \mathbb{R}$.

$$\lambda . \vec{A} \triangleq (\lambda a_1, \lambda a_2, \dots, \lambda a_n).$$

OBS.: SI $\vec{A} \neq \vec{0}$ Y $\lambda \neq 0$, ENTONCES $\lambda . \vec{A}$ ES UN VECTOR QUE TIENE LA MISMA DIRECCIÓN QUE $\vec{A}$ (PARALELO)

- $\lambda > 0 \Rightarrow \vec{A}$ Y $\lambda \vec{A}$ TIENEN IGUAL SENTIDO
- $\lambda < 0 \Rightarrow \vec{A}$ Y $\lambda \vec{A}$ TIENEN SENTIDOS OPUESTOS

EJ: $\vec{A} = (2, 1, -3) \in \mathbb{R}^3$

$$-5 . \vec{A} = -5 . (2, 1, -3) = (-10, -5, 15)$$

PROPIEDADES:

SI $\vec{A}, \vec{B}, \vec{C} \in \mathbb{R}^n$ Y $\lambda, \mu \in \mathbb{R}$, ENTONCES

1) $A + (B+C) = (A+B) + C$

2) $A + B = B + A$.

3) $\lambda(\vec{A} + \vec{B}) = \lambda\vec{A} + \lambda\vec{B}$

4) $(\lambda+\mu).\vec{A} = \lambda\vec{A} + \mu\vec{A}$

5) $(\lambda.\mu).\vec{A} = \lambda(\mu.\vec{A})$

6) $\vec{0} + \vec{A} = \vec{A}$

7) $1.\vec{A} = \vec{A}$

8) $\vec{A} + (-1)\vec{A} = \vec{0} \longrightarrow -\vec{A} \stackrel{d}{=} -1.\vec{A}$

$\vec{A} + (-1).\vec{A} = 1\vec{A} + (-1).\vec{A} =$

$= [1 + (-1)].\vec{A} = 0.\vec{A} = \vec{0}$

9) $0.\vec{A} = \vec{0}$

OBS: $-\vec{A} = (-1).\vec{A}$

$\vec{A} + (-\vec{A}) = \vec{0}$

EJ: $\vec{A} = (-5, 2, 7)$, $-\vec{A} = (5, -2, -7)$

$\vec{A} + (-\vec{A}) = (0, 0, 0)$

OBS/DEF: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

EJ: $\vec{A} = (2,-1,3)$, $\vec{B} = (-1,2,-4)$

$3\vec{A} - \vec{B} = 3(2,-1,3) - (-1,2,-4) =$

$= (6,-3,9) - (-1,2,-4) = (7,-5,13)$

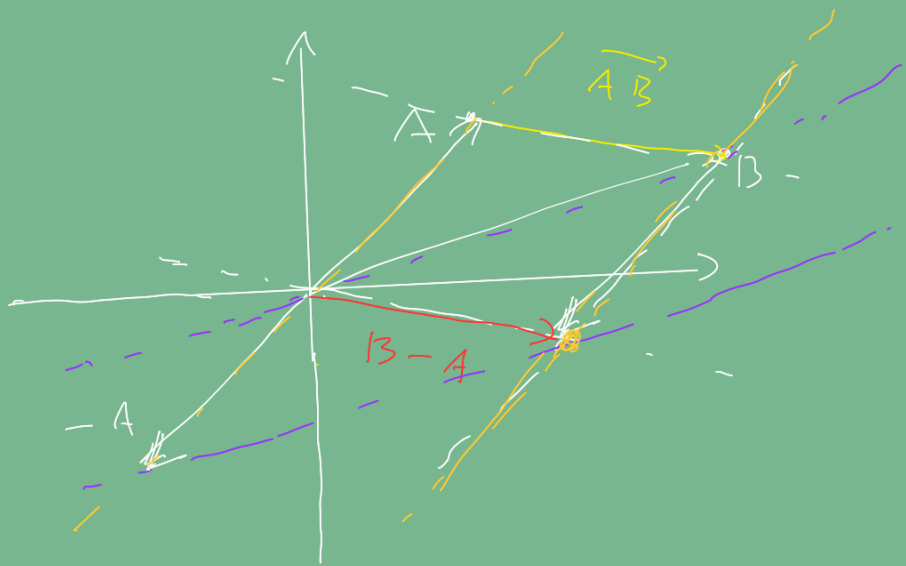
$\vec{A} + 2\vec{B} = (2,-1,3) + 2(-1,2,-4) =$

$= (2,-1,3) + (-2,4,-8) = (0,3,-5)$

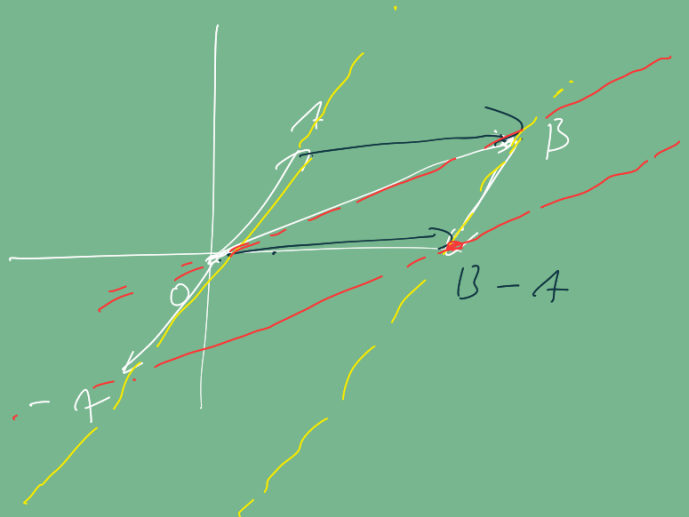
ACLARACIÓN: SI $A = (a_1, a_2, \dots, a_m)$ Y $B = (b_1, b_2, \dots, b_m)$, ENTONCES

$A = B \iff a_1 = b_1, a_2 = b_2, \dots, a_m = b_m$.

RESTA DE VECTORES:



$B - A = B + (-A)$



Ej: $A = (3,5)$; $B = (4,1)$

$B - A = (4-3, 1-5) = (1, -4)$

